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MONGO -DB CRUD OPERATION:

```
import com.mongodb.client.FindIterable;
import com.mongodb.client.MongoClient;
import com.mongodb.client.MongoClients;
import
com.mongodb.client.MongoCollection;
import com.mongodb.client.MongoDatabase;
import com.mongodb.client.model.Filters;
import com.mongodb.client.model.Updates;
import org.bson.Document; public class Test2
{   public static void main(String[] args) {
        // Connect to MongoDB
        MongoClient mongoClient =
MongoClients.create("mongodb://localhost:27017");
        // Access Database and Collection
        MongoDatabase database = mongoClient.getDatabase("CollegeDB");
        MongoCollection<Document> collection = database.getCollection("Students");
        // =====
        // CREATE (Insert 5 Students)
        // =====
collection.insertOne(new Document("id", 101)
        .append("name", "Aarav")
        .append("department", "CSE")
.append("cgpa", 8.5));    collection.insertOne(new
Document("id", 102)
        .append("name", "Priya")
        .append("department", "IT")
.append("cgpa", 9.1));    collection.insertOne(new
Document("id", 103)
        .append("name", "Rahul")
```

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```
.append("department", "ECE")

.append("cgpa", 8.0));    collection.insertOne(new
Document("id", 104)

.append("name", "Sneha")

.append("department", "ME")

.append("cgpa", 8.8));

collection.insertOne(new Document("id", 105)

.append("name", "Vikram")

.append("department", "CIVIL")

.append("cgpa", 7.9));

System.out.println("5 Documents Inserted Successfully");

// =====

// READ (Display All Students)

// =====

System.out.println("\nStudent Records:");

FindIterable<Document> students = collection.find();    for
(Document doc : students) {

    System.out.println(doc.toJson());

}

// =====

// UPDATE

// Update CGPA of Student ID 103

// =====

collection.updateOne(

    Filters.eq("id", 103),

    Updates.set("cgpa", 9.3)

);

System.out.println("\nStudent ID 103 Updated Successfully");
```

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```
// =====  
// READ Updated Record  
// =====  
  
Document updatedStudent =  
collection.find(Filters.eq("id", 103)).first();  
System.out.println("\nUpdated Record:");    if  
(updatedStudent != null) {  
    System.out.println(updatedStudent.toJson());  
}  
// =====  
// DELETE  
// Delete Student ID 105  
// =====  
collection.deleteOne(Filters.eq("id", 105));  
System.out.println("\nStudent ID 105 Deleted Successfully");  
// =====  
// DISPLAY FINAL RECORDS  
// =====  
System.out.println("\nFinal Student Records:");  
for (Document doc : collection.find()) {  
    System.out.println(doc.toJson());  
}  
// Close Connection  
mongoClient.close();  
}
```

For Many:

```
package mongodb2; import  
com.mongodb.client.FindIterable; import
```

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```
com.mongodb.client.MongoClient; import
com.mongodb.client.MongoClients; import
com.mongodb.client.MongoCollection;
import com.mongodb.client.MongoDatabase;
import com.mongodb.client.model.Filters;
import com.mongodb.client.model.Updates;
import java.util.Arrays; import
org.bson.Document; public class Test2 {
public static void main(String[] args) {
    // Connect to MongoDB

    MongoClient mongoClient =
MongoClients.create("mongodb://localhost:27017");

    // Access Database and Collection

    MongoDatabase database = mongoClient.getDatabase("CollegeDB");

    MongoCollection<Document> collection = database.getCollection("Students")

    // CREATE (Insert 5 Students)
collection.insertMany(Arrays.asList(
new Document("id", 101)
    .append("name", "Aarav")
    .append("department", "CSE")
    .append("cgpa", 8.5),
new Document("id", 102)
    .append("name", "Priya")
    .append("department", "IT")
    .append("cgpa", 9.1),
new
Document("id", 103)
    .append("name", "Rahul")
    .append("department", "CSE")
    .append("cgpa", 8.0),
new
Document("id", 104)
```

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```
        .append("name", "Sneha")
        .append("department", "ME")
.append("cgpa", 8.8),          new
Document("id", 105)

        .append("name", "Vikram")
        .append("department", "CSE")
        .append("cgpa", 7.9)));

System.out.println("5 Documents Inserted Successfully");

// READ (Display All Students)
System.out.println("\nStudent Records:");
for (Document doc : collection.find()) {
    System.out.println(doc.toJson());
}

// UPDATE
collection.updateMany(
    Filters.eq("department", "CSE"),
    Updates.set("department", "Computer Science")
);

System.out.println("\nMultiple Documents Updated Successfully");    //
READ Updated Record

System.out.println("\nRecords After Update:");
for (Document doc : collection.find()) {
    System.out.println(doc.toJson());
}

// Delete Many
collection.deleteMany(
    Filters.eq("department", "Computer Science")
);

System.out.println("\nMultiple Documents Deleted Successfully");

// DISPLAY FINAL RECORDS
```

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```
        System.out.println("\nFinal Student Records:");  
for (Document doc : collection.find()) {  
    System.out.println(doc.toJson());  
}  
    // Close Connection  
mongoClient.close();  
}  
}
```